

Brahim EL MOKHTARI

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Education

- 2022 – 2025

 **Ph.D. in Electrical, Electronic and Systems Engineering.**
Pascal Institute, Université Clermont Auvergne, France
Thesis title: *Source Identification Approaches for EMC and SI Issues: Opening Towards Machine Learning Models*
- 2021 – 2022

 **Master 1 (First-Year M.Sc.) in Electronics, Electrical Energy, and Automatics (EEEA),** specializing in Electromagnetic Compatibility (EMC).
EUPI, Université Clermont Auvergne, France.
- 2020 – 2020

 **Virtual Reality Training (EON XR Developer Level 2).** Mohammed VI University - IDC Morocco
- 2018 – 2021

 **Ph.D. in Nuclear Physics.** Faculty of Sciences, Mohammed V University, Rabat, Morocco.
- 2015 – 2017

 **Master's Degree in Nuclear Physics and Radiation Technologies.** Faculty of Sciences, Mohammed V University, Rabat, Morocco.
- 2015 – 2016

 **Bachelor of Science (B.Sc.) in Fundamental Physics.** Lille 1 University - Science and Technology, France

Professional Experience

- 2026 (6 months)

 **Temporary Assistant Professor / Research and Teaching Fellow (ATER),** Université Clermont Auvergne, France.
- 2023 (3 months)

 **Co-supervisor for M2 Master's Mini-Projects (EMC),** Pascal Institute, UCA, France.
- 2022 (3 months)

 **Master 1 (M1) Internship:** Development of a graphical user interface (GUI) for EMC simulations.
Pascal Institute, UCA, France.
- 2020 (6 months)

 **VR Developer:** Development of a mobile VR application "Virtual Fitness Coach". IDC Morocco, UM6P.
- 2017 (5 months)

 **Master 2 (M2) Internship:** Modeling and simulation of radiation-induced effects in the collimator of the PGAA facility within the NB1 channel of the TRIGA Mark II reactor. CNESTEN, Morocco.

Research Publications

Journal Articles

- 1

B. El Mokhtari, C. Chauviere, and P. Bonnet, "On the importance of the mathematical formulation to get pinns working," *IEEE Transactions on Electromagnetic Compatibility*, vol. 66, no. 6, pp. 2142–2149, 2024.  DOI: 10.1109/TEMPC.2024.3490699.
- 2

B. El Mokhtari, C. Jullien, and P. Bonnet, "Active suppression of transient crosstalk disturbances in transmission line networks," *IEEE Transactions on Electromagnetic Compatibility*, vol. 66, no. 6, pp. 1900–1908, 2024.  DOI: 10.1109/TEMPC.2024.3481632.
- 3

L. El Amri et al., "New open-source software for gamma-ray spectra analysis," *Applied Radiation and Isotopes*, vol. 185, p. 110 227, 2022.  DOI: 10.1016/j.apradiso.2022.110227.
- 4

L. El Amri et al., "Neutron guide optimization for the moroccan pgaa system," *Applied Radiation and Isotopes*, vol. 174, p. 109 783, 2021.  DOI: 10.1016/j.apradiso.2021.109783.
- 5

B. El Mokhtari et al., "Modeling and simulating the induced effect on the steel collimator plug used in the pgaa facility of the moroccan triga mark-ii reactor under different neutron irradiation levels," *Applied Radiation and Isotopes*, vol. 170, p. 109 620, 2021.  DOI: 10.1016/j.apradiso.2021.109620.

Conference Proceedings with Oral Presentation

- 1

B. El Mokhtari and P. Bonnet, "Software correction of transient crosstalk interferences in unshielded wiring: Numerical and experimental validations," in *Proceedings of the The 46th PIERS 2025 in Abu Dhabi*, 2025.
- 2

B. El Mokhtari, P. Bonnet, G. Al Achkar, and C. Jullien, "Supression active des signaux de perturbation dans les lignes de transmission," in *Proceedings of the The 21th Colloque International et Exposition sur la Compatibilité ÉlectroMagnétique*, 2023.
- 3

B. El Mokhtari and P. Bonnet, "Alternative active solution for crosstalk reduction.," in *2023 International Conference on Electromagnetics in Advanced Applications (ICEAA)*, 2023, pp. 611–611.  DOI: 10.1109/ICEAA57318.2023.10297929.

Skills

- Languages

 Fluent/Proficient in English, French, and Arabic (Reading, Writing, and Speaking).
- Coding

 Python, TensorFlow, Keras, PyTorch, MATLAB, Java, Java Swing, Fortran, C, C#, LaTeX, ...
- Software & Tools

 CST Studio Suite, COMSOL Multiphysics, LTspice, MCNP, 3ds Max, Unity, Linux, ...

Interests

Traveling, Fitness, Tennis, Swimming, ...



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Article

Lead-free Cs₂AgV(Se)₄ kesterite absorber for high-efficiency thin-film solar cells investigations: Integrated first-principles, device simulation, and AI predictions

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Citations

Elkhoul, A., El Badraoui, Abdelmajid, El Mokhtari, Brahim, ... Drissi, Lalla Btissam, Tahiri, Najim

Solar Energy Materials and Solar Cells, 2026, 306, 114563

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Article

Multifunctional lead-free double perovskites Cs₂CuScX₆ (X = Br, I) for photovoltaic and thermoelectric applications: A DFT and SCAPS-1D study

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Citations

El Badraoui, Abdelmajid, Elkhoul, A., El Mokhtari, Brahim, ... Ez-Zahraouy, Hamid, Tahiri, Najim

Materials Chemistry and Physics, 2026, 358, 132423

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Article

Optimization of lead-free CsSnBr₃-based perovskite solar cells via SCAPS-1D simulations and machine learning models

El Badraoui, Abdelmajid, Chargui, Taoufik, Elkhoul, A., ... Ez-Zahraoui, Hamid, Tahiri, Najim

Journal of Physics and Chemistry of Solids, 2026, 210, 113340

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Citations

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Article

On the Importance of the Mathematical Formulation to Get PINNs Working

El Mokhtari, Brahim, Chauvière, Cédric, Bonnet, Pierre

IEEE Transactions on Electromagnetic Compatibility, 2024, 66(6), pp. 2142–2149

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Citations

Article

Active Suppression of Transient Crosstalk Disturbances in Transmission Line Networks

El Mokhtari, Brahim, Jullien, Charles, Bonnet, Pierre

IEEE Transactions on Electromagnetic Compatibility, 2024, 66(6), pp. 1900–1908

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Citations

Conference Paper

Alternative Active Solution for Crosstalk Reduction

El Mokhtari, Brahim, Bonnet, Pierre

2023 International Conference on Electromagnetics in Advanced Applications, ICEAA 2023, 2023, pp. 611

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Citations

Review

New open-source software for gamma-ray spectra analysis

El Amri, Lahcen, Chetaine, Abdelouahed, Amsil, Hamid, ... Laraki, Khalid, Marah, Hamid

Applied Radiation and Isotopes, 2022, 185, 110227

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Citations

Article

Neutron guide optimization for the Moroccan PGAA system

El Amri, Lahcen, Chetaine, Abdelouahed, Amsil, Hamid, ... Laraki, Khalid, Marah, Hamid

Applied Radiation and Isotopes, 2021, 174, 109783

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Article

Modeling and simulating the induced effect on the steel collimator plug used in the PGAA facility of the Moroccan TRIGA Mark-II reactor under different neutron irradiation levels

12 Citations

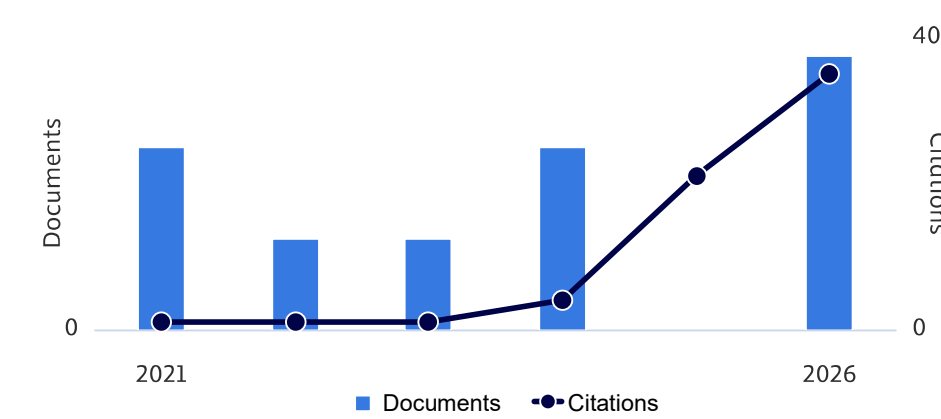
El Mokhtari, Brahim,Chetaïne, Abdelouahed,Amsil, Hamid,... Laraki, Khalid,Marah, Hamid
Applied Radiation and Isotopes, 2021, 170, 109620

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